# Brandon Jahoor

*Robotics Software Engineering for Humanoid Embodiment of Physical AI*

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**University of Waterloo** | *Candidate for BASc in Mechatronics Engineering Honours, Minor in Artificial Intelligence* *2028*

## Skills

**Robotics**: Isaac Sim & Lab, URDF/USD, ROS2, controls, IK/FK, Nav2, path planning
**ML & Perception**: ACT, VLA, RL, LeRobot, Hugging Face, computer vision, OpenCV, RealSense
**Software**: Python, C++, Docker, Linux/Ubuntu, Claude Code, llama.cpp, Ollama
**Hardware & Design**: SolidWorks, 3D printing, SO-ARM101, Nvidia Jetson, Raspberry Pi, prototyping

## Experience

**Robotics Software Engineering (Co-op)** Sept 2026 – April 2027
*Axibo AI* | *Humanoids* | *site* · *portfolio* *Ontario*

- Fine-tune, evaluate and deploy SOTA VLAs (**$\pi_{0.5}$**, **GR00T N1.7**, **DM0.5**) end-to-end as generalizable bimanual and whole-body policies, both integrated with the **WBC** and locomotion policies, with a focus on cross-embodiment and sim-to-real.
- Created and designed a batched synthetic data generation toolchain for delta-EE bimanual policies using **Genesis World**, **Isaac**, **MuJoCo** and **cuRobo** (motion planning). Features like augmentation, domain-randomization, noise improve **cross-embodiment**.
- Accelerated SOTA VLA research iteration and cut compute cost with custom training pipelines and scripts forked from SOTA repos, primarily **openpi** (**JAX/Orbax**) (**2.5**× faster steps) and **Isaac-GR00T** (**PyTorch**), without sacrificing benchmarks.
- Optimized VLA training through sharding across multi-datacenter GPUs, upgraded version pins, selective weight and optimizer-state quantization, partial backbone freezing/pruning, selective rematerialization and batched checkpointing.
- Built batched sim evaluation toolchains with custom success criteria (per-component scoring with penalties) for SOTA bimanual & whole-body policies, ran **sim-to-sim** & **sim-to-real** evaluation on a variety of humanoids & bimanual arms, and raised benchmark scores by tuning training recipes & inference settings.
- Optimizing real-time inference via **RTC** & Mixture of Horizons (**MoH**), tuning denoising steps & prediction horizons.
- Productionizing the custom **UMI** pipeline and developed training recipes to maximize the efficacy of public data and synthetically generated data (including re-simulating public datasets with custom modifications) while minimizing required real-world data.
- Deploy bimanual and whole-body policies end-to-end, developing the deployment tooling, workflows and runtime as forks of SOTA systems, integrating **WBC** and **RTC**.
- Develop the team's **agent-centric** pipelines, workflows and tooling across the data, training and evaluation stack.
- Architect humanoid runtimes spanning **LLM/agent-harness** orchestration of VLA skills, VLA-driven whole-body control and manipulation policies, and an upper-body VLA with a lower-body locomotion policy, all coordinated through the **WBC**.

**Robotics Software Engineering (Co-op)** Jan 2026 – April 2026
*Thalassa Robotics* | *Subsea Robotics* | *site* · *portfolio* *California*

- Built a high-fidelity toolchain on **Isaac Sim** used team-wide, integrating and configuring **optics and hydrodynamics** it lacks.
- Prepared the robot's **USD** model and joint articulation so the simulated assembly matched the real one.
- Layered control abstractions so simulated motors drive exactly like the real servos, with acceleration-bounded interpolation.

**Robotics Software Engineering (Research Assistant)** Sept 2025 – Dec 2025
*University of Waterloo RoboHub* | *Robotics Research Lab* | *gitlab* · *site* · *portfolio* *Ontario*

- Streamed **human arm motion** from **Xsens MOCAP** into **ROS2** in real time, for retargeting onto a 7-axis **Franka Emika Panda**.
- Containerised the **ROS2** stack in **Docker** so results reproduced across the lab's workstations.

**Robotics Engineering (Co-op)** June 2025 – Aug 2025
*University of Waterloo Robotics Team* | *Mars Rover Team* | *github* · *site* · *portfolio* *Ontario*

- Developed a **ROS2 RealSense RGB-D** pipeline and solved **Mars rover drivetrain kinematics** on an **Nvidia Jetson**.

**Robotics Engineering (Co-op)** May 2025 – June 2025
*BHF Robotics* | *Agricultural Robotics* | *site* · *portfolio* *Ontario*

- Implemented **ROS2 autodrive** and trained a **deep-learning** weed/crop detector for an autonomous farming robot.

## Projects

**VLA Finetuning** | *SmolVLA, LeRobot, SO-ARM101, Nvidia Jetson* | github · portfolio

- Trained **ACT** and **SmolVLA** on 50–100 teleoperated demos/task, running **30 Hz** on a Jetson Orin Nano.
- Optimized batch and worker counts for **LeRobot PyTorch** training on an **RTX 3060 Ti**.

**Chess Robot** | *SO-ARM101, SolidWorks, 3D printing, LeRobot, Placo, Python* | *Stealth Startup* | portfolio

- **CAD**-modified an **SO-ARM101** and built its rig, then developed a three-call Python API that plugs straight into a chess engine.
- Extended reach and fused the wrist to drop a motor, updating the **URDF** model, solver constraints and **LeRobot** driver.
- Built the motion engine over **LeRobot** and **Placo** IK: convergence loops, **1 mm** waypoints, **PID gravity compensation**.

**Autonomous Navigation Stack** | *ROS2, C++, LiDAR, A*, Foxglove* | *WATonomous* | github · portfolio
**Robot Arm Simulation** | *Isaac Sim, Isaac Lab, CUDA, RL* | *Stealth Startup* | portfolio
**Hexapod Robot** | *Nvidia Jetson, OWL-ViT, SO-ARM101, RealSense* | portfolio
**AI Roommate** | *Nvidia Jetson, Qwen2.5-0.5B, YOLOv8s, FastAPI* | github · portfolio
**Autonomous Rover Simulation** | *Gazebo, ROS2, Nav2, Python* | *RoboHub* | github · portfolio
**AI Object Detection** | *ROS2, OWL-ViT, RealSense, OpenCV, Python* | github · portfolio

***See my portfolio: bjahoor.github.io***